INSTRUCTION MANUAL
PoliKick

Analog Eurorack KickDrum developed by the student team PoliTeK - Politecnico di Torino

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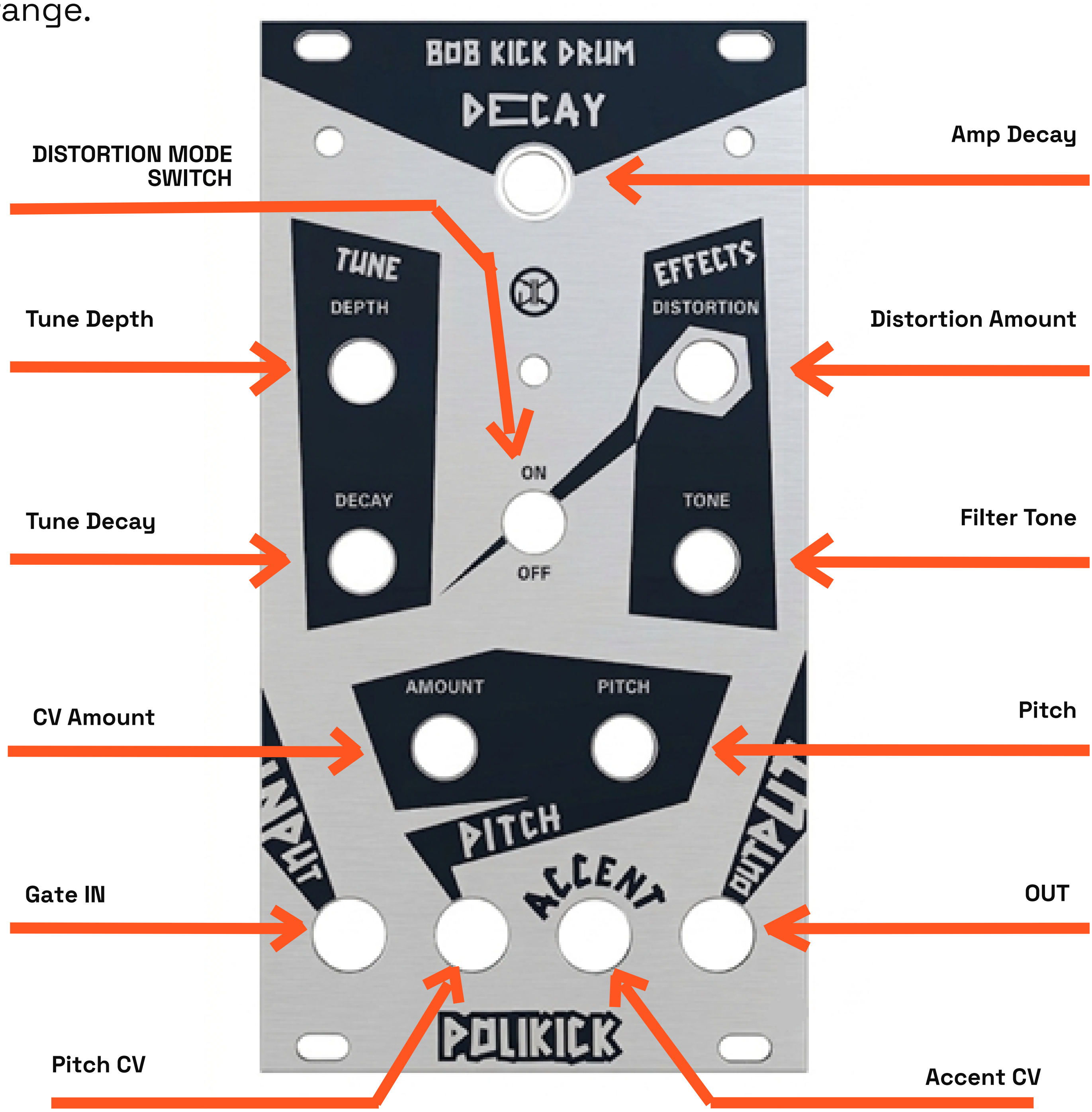
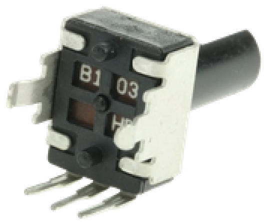
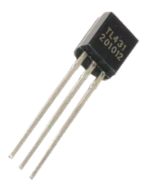
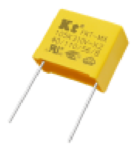
INTRODUCTION

The PoliKick is an analog kick drum designed for integration into a Eurorack modular system. Inspired by the iconic Roland TR-808, the module adapts this historical design to the requirements and possibilities of the modular format. The circuit draws inspiration from the EDU DIY kick drum by Erica Synths, introducing an additional selectable distortion stage and an extended pitch range.



HARDWARE AND MATERIALS

The components required to build the PoliKick include very common resistor values (E12 series), transistors, op-amps, and capacitors. You will likely already have most of these components available in your laboratory, but the complete Bill of Materials (BOM) can be found in our GitHub repository (https://github.com/PoliTeK/PoliKicK).
These components will be soldered onto the PCB. You can find the manufacturing print files in a dedicated folder within the repository to have them manufactured. Using the files provided in the same repository, it is also possible to have the front panel manufactured in FR-4 (the same material used for printed circuit boards). This solution allows you to obtain a solid, professional front panel while producing both the board and the front interface in a single order.


Electronic Components

PCB

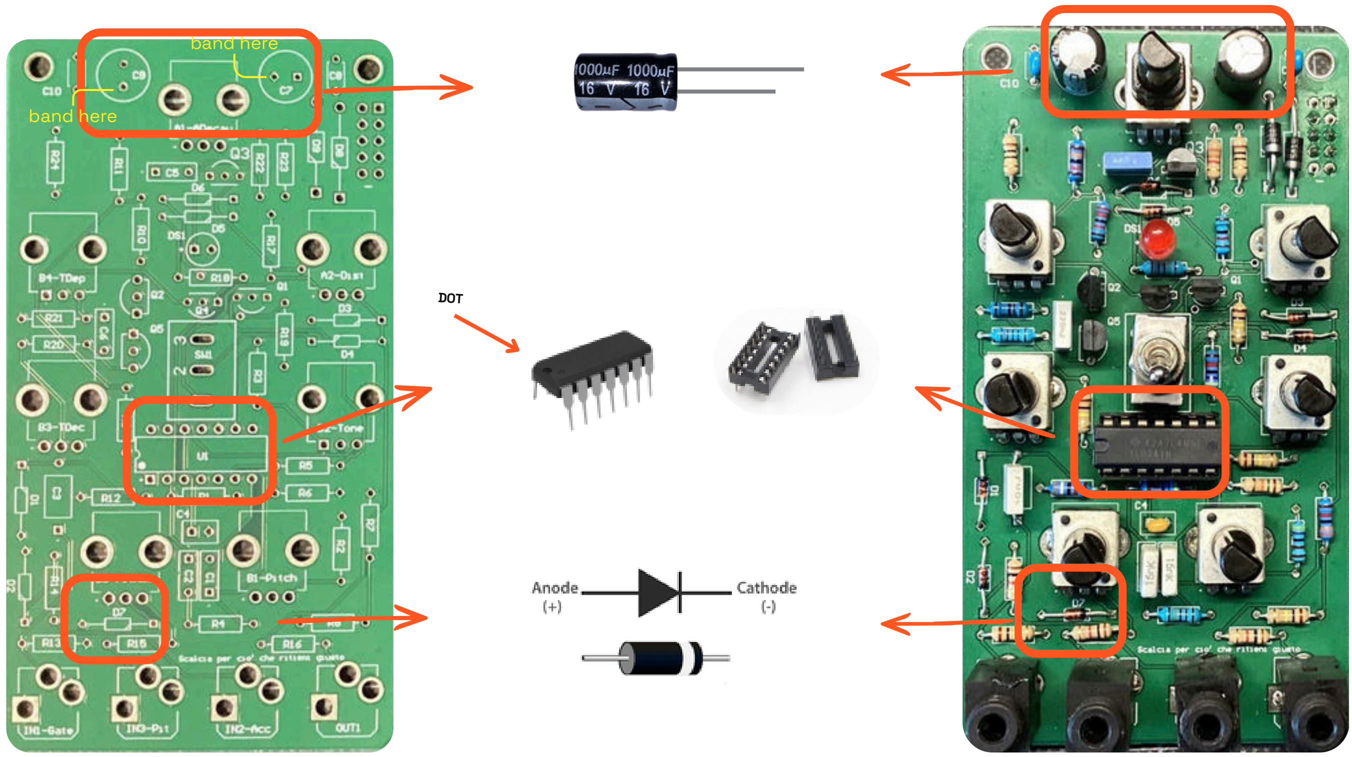
M3 SCREWS AND NUTS 1.5CM STANDOFFS

FRONT PANEL

ASSEMBLY

First, assemble the PCB by soldering all electronic components. It is recommended to solder the smaller components first, such as resistors and capacitors, and solder the connectors, potentiometers, and the switch only at the very end.
The power connector must be soldered onto the back of the board, meaning the opposite side relative to all other components.
During assembly, pay close attention to polarity: diodes, electrolytic capacitors, and the LED have a mandatory insertion direction.

- Align the cathode band of the diodes exactly as drawn on the PCB.
- Align the side stripe of the electrolytic capacitors with the round pad.
- Align the dots on the Op-Amp; solder the sockets first, and then insert the Op-Amp into them.
- Align the leads of the LED as drawn on the PCB (longer lead to the +).
- The red power LED should be soldered last so it can be easily aligned with the front panel hole.



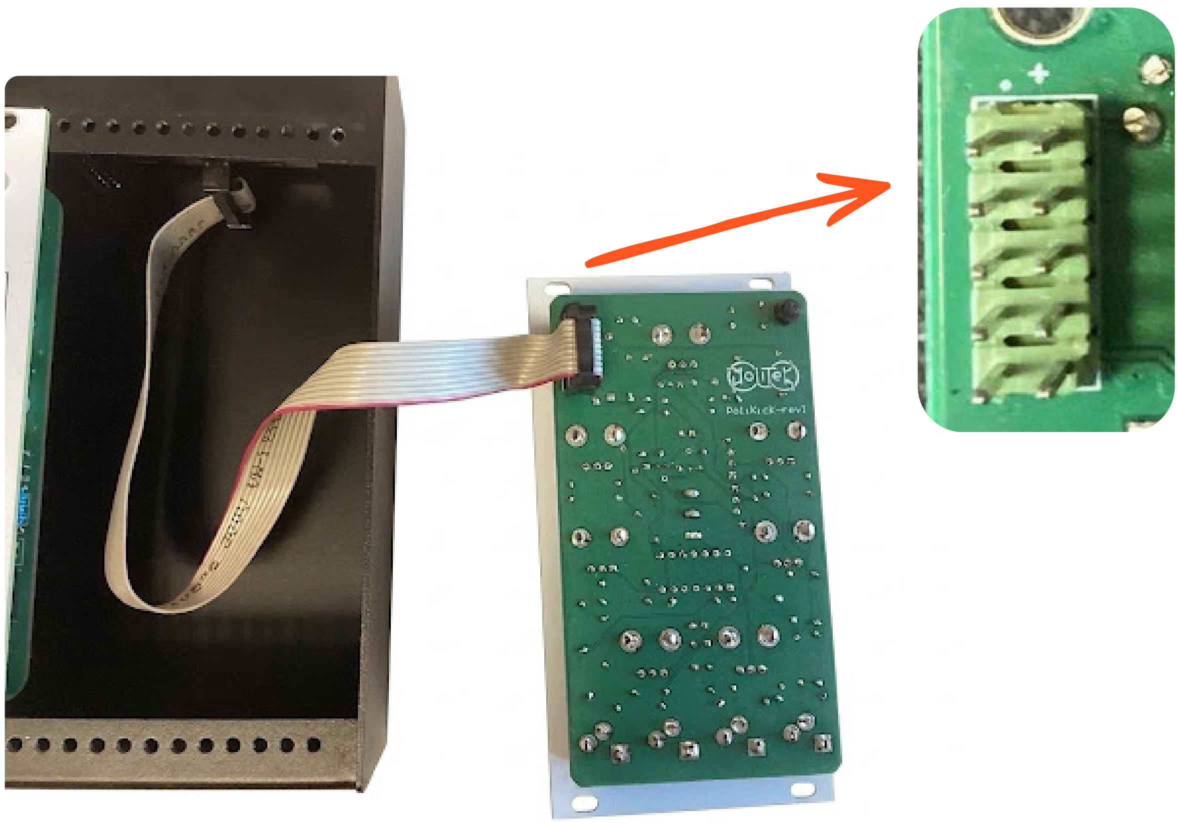
ASSEMBLY

Once the PCB soldering is complete, it is time to attach the board to the front panel. Place the panel over the circuit, ensuring that the holes are perfectly aligned with the potentiometers, jacks, switch, and LED. To secure everything firmly, screw the nuts for the audio jacks and the switch into their respective positions. Finally, use the screws, nuts, and standoffs in the designated PCB holes to tighten the structure and ensure maximum mechanical stability for the module. To conclude the assembly, push the potentiometer knobs onto the shafts.

ASSEMBLY

The PoliKick is now complete and ready to be mounted into your Eurorack case; follow these steps for a safe installation:

- Turn off the power: Before making any connections, ensure that the case power supply is turned completely off and disconnected from the mains electricity.
- Connect the ribbon cable: To power the module, you must use a standard ribbon cable. Plug it into the connector on the back of the PCB, paying strict attention to the polarity by aligning the red stripe of the cable with the "-" indicator printed on the silkscreen.
- Connect to the bus board: Insert the opposite end of the ribbon cable into the power distribution bus board of your case, verifying once again that the red stripe is correctly oriented toward the -12V rail.
- Secure the module: Position the PoliKick into the desired slot and screw the front panel onto the rails using the appropriate mounting screws, taking care not to overtighten them to avoid scratching the FR-4 surface.
- Power on: Supply power to the system, patch an input signal into the Trigger jack, and turn up the levels to start using your module.



TECHNICAL DOCUMENTATION

If you want to learn more or dive deeper into the circuit theory behind this project, you can download our comprehensive technical documentation, where you will find the complete circuit analysis, including all formulas and detailed calculations! Feel free to contact us with any questions, doubts, or if you would like to propose modifications and improvements to the project!

Telegram Community

Repository GitHub

Tecnical Documentation